

TALKING CLIMATE: DOES PEER ANXIETY AMPLIFY OR ALLEVIATE CLIMATE
CHANGE WORRIES?

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Talking Climate: Does Peer Anxiety Amplify or Alleviate Climate Change Worries

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Abstract

As the climate crisis intensifies, climate change anxiety (CCA) has become increasingly common, yet it remains unclear whether discussing climate concerns with peers alleviates or amplifies distress. This study examined how climate conversations related to daily climate change anxiety, optimism, and pro-environmental behavior (PEB), as well as whether the perceived emotional tone of conversation partners shapes these associations. Using an 8-day daily diary design (N = 121), participants reported their daily climate-related emotions, engagement in private and public PEBs, and perceptions of peers' climate anxiety and optimism following climate-related conversations. Results indicated that engaging in climate conversations was associated with higher daily climate anxiety and greater private PEB, but was not significantly associated with optimism or public PEB. Individuals' own anxiety and optimism were both positively associated with private and public PEB, and optimism strengthened the relationship between anxiety and public behavior. Mediation analyses further showed that CCA explained the association between climate conversations and private PEB. Perceived peer emotional tone was not significantly associated with participants' own emotions or behaviors. These findings suggest that climate conversations may increase emotional engagement, which can motivate action, particularly at the individual level.

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Does Peer Anxiety Amplify or Alleviate Climate Change Worries?

Climate change is not just an environmental crisis—it is an emotional one, shaping the way individuals think, feel, and interact with others. Climate change is one of the most pressing global challenges of the 21st century, with widespread environmental, economic, and psychological consequences (Intergovernmental Panel on Climate Change, 2021; American Psychological Association, 2017).. Climate change anxiety (CCA) is a distressing emotional response to the perceived threat of ecological crisis (Clayton & Manning, 2018). Understanding how individuals cope with CCA is crucial for learning how to face ecological issues while promoting pro-environmental behaviors (PEB) to effectively manage the climate crisis. Conversations about climate change with close others (e.g., immediate and extended family members, romantic partners, friends, classmates, coworkers, neighbors, online contacts) may play an important factor in shaping climate-related emotions and ensuing PEB, but their unique and interactive effects remain unclear. Research grounded in stress-buffering theory suggests that supportive social interactions can mitigate the negative psychological effects of stress (Cohen & Willis, 1985), whereas emotion contagion theory proposes that individuals may “catch” the emotions expressed by others during social exchanges (Hatfield et al., 1993). Accordingly, social interactions may serve as both a buffer against and a source of climate-related distress, depending on the emotional tone of the discussion (Pearce et al., 2020). In particular, the emotional expressions of conversation partners may shape whether one’s own climate-related anxiety becomes motivating or overwhelming. Despite growing research on social influences in climate engagement, little is known about how the emotional outlook of a peer shapes an individual’s climate anxiety and subsequent behavior, particularly in the context of PEB, including both private actions and public actions (Stevenson et al., 2019; Tian et al., 2025).

Understanding how social interactions impact CCA and climate-related behavioral engagement is essential for developing effective interventions that promote resilience and collective action.

Although decades of research on social influence processes, including Social Identity Theory and emotional contagion, demonstrate that social interactions shape emotions and behavior (Tajfel & Turner, 1979; Hatfield et al., 1993), CCA may represent a distinct case. Unlike many everyday stressors, climate change is a large-scale, abstract, and collective threat that is temporally distant yet personally relevant (Reser et al., 2012; Clayton & Manning, 2018). As a result, climate-related emotions may be more ambiguous, difficult to regulate, and less clearly tied to immediate behavioral outcomes (Ojala, 2016, 2022). This may alter how social interactions influence emotional responses and engagement, making it important to examine these processes specifically within the context of climate change.

This study explores whether interpersonal conversations influence CCA and PEB, and how the emotional tone of these discussions, particularly the conversation partner's levels of climate-related anxiety and optimism, may moderate this relationship. Stress and optimism play complex roles in motivation. While stress can motivate action when managed effectively, it may also lead to helplessness when it becomes overwhelming. In psychological research, this distinction is often conceptualized as eustress, which can facilitate adaptive coping and engagement, versus distress, which may impair functioning and lead to avoidance (Ojala & Bengtsson, 2019). In the context of climate change, moderate levels of anxiety may promote engagement, whereas excessive anxiety may contribute to disengagement.

Alternatively, optimism can encourage engagement and problem-solving, making it an important factor in whether climate-related discussions are associated with PEB. However, excessive optimism may also lead to complacency, as individuals may feel that the problem will

be resolved without their direct involvement (Ojala & Lakew, 2017). In the present study, PEB refers to self-reported private (e.g., reducing consumption) and public (e.g., advocacy or awareness-raising) behaviors, which are conceptualized as indicators of psychological and behavioral engagement with climate change. While these behaviors may vary in their direct environmental impact, they are widely used in psychological research as markers of engagement and intent (Stevenson et al., 2019).

Given that emotional responses to climate discussions can shape individuals' perceptions of their ability to make a difference and their likelihood of engaging in climate-related behaviors, it is important to examine how these conversations influence both CCA and behavioral engagement. This study is not intended to evaluate the objective environmental impact of these behaviors, but rather to examine the psychological processes that link social interaction, emotional responses, and self-reported engagement in the context of climate change.

Climate Change and Its Emotional Consequences

Climate change is associated with a range of environmental, economic, and psychological challengers, with potential implications for both human and ecological well-being (Leiserowitz et al., 2021). Beyond environmental impacts, climate change has been linked to psychological responses such as concern, worry, and climate-related distress. Recent work in ecopsychology and ecopolitical psychology has conceptualized CCA as an emotional response to perceived environmental threats, which may reflect both adaptive concern and distress depending on its intensity and context (Clayton & Manning, 2018; Pihkala, 2020). The widespread psychological effects of climate change further intensify the crisis, with individuals experiencing heightened distress, eco-anxiety, and even trauma in response to environmental degradation (Latkin et al., 2024).

Importantly, climate-related emotional responses are not inherently maladaptive. Moderate levels of concern may promote engagement and problem-solving, whereas more intense or chronic distress may be associated with feelings of helplessness or disengagement (Ojala, 2016). In this context, CCA is examined not as a clinical disorder, but as a continuum of emotional responses that may have varying implications for psychological well-being and behavior. Although some research has suggested that climate-related distress may be experienced as highly salient or even overwhelming, the present study does not treat such responses as equivalent to clinical trauma. Rather, the focus is on everyday emotional experiences related to climate change and how these responses are associated with behavioral engagement. Given these considerations, the present study does not aim to evaluate the objective of climate change or to promote specific behavioral outcomes. Instead, it seeks to examine how climate-related emotions and social interactions are associated with self-reported engagement in PEBs, and how these relationships unfold in daily life.

CCA has received increasing attention in recent years, particularly among young people, although the extent to which it represents a primary concern varies across individuals and contexts (Ojala & Lakew, 2017). CCA encompasses feelings of worry, fear, and uncertainty regarding environmental threats, and while some degree of anxiety can be adaptive by motivating individuals to take action, excessive distress may lead to disengagement and mental health challenges.

Ojala and Bengtsson (2019) found that adolescents who lacked coping mechanisms or emotional support from peers and family were more likely to experience immobilizing distress—a form of emotional overwhelm marked by withdrawal, avoidance, and a sense of helplessness that undermines motivation to engage in climate-related behaviors. These findings suggest that

CCA can function as either a motivator or a barrier to engagement, depending on whether individuals possess adaptive coping resources and emotional support.

At the same time, emerging research suggests that heightened and persistent climate-related worry may also be associated with lower well-being and reduced life satisfaction in some individuals (Verplanken et al., 2020). This highlights the importance of examining not only when climate-related concern promotes engagement, but also when it may contribute to distress or disengagement. As such, the present study focuses on understanding the conditions under which climate-related emotions are associated with adaptive versus maladaptive outcomes in daily life.

Coping with Climate Change and Its Emotional Consequences

There are various coping mechanisms to help alleviate CCA and sustain engagement in climate action, including cognitive reframing, emotional regulation, and social support (Ojala & Bengtsson, 2019; Pearce et al., 2020). Importantly, coping strategies not only shape emotional well-being but also influence whether climate-related distress translates into PEB.

Cognitive reframing, a cognitive-behavioral therapy technique, helps individuals shift their perspective on climate change from an overwhelming crisis to a challenge with actionable solutions, thereby enhancing resilience and motivation to act (Ojala & Bengtsson, 2019). Similarly, emotion regulation, the ability to manage distress while maintaining goal-directed behavior, plays a critical role in sustaining climate engagement (Pearce et al., 2020). Individuals who adopt adaptive regulation strategies, such as acceptance, reappraisal, and problem-focused coping, are more likely to translate climate concern into sustained PEB, including conservation efforts and environmental advocacy (Armstrong et al., 2025; Ojala & Bengtsson, 2019). In contrast, unregulated or overwhelming distress is associated with avoidance and disengagement.

Coping processes may also be supported through social interactions. For example, adolescents who engaged in constructive discussions about climate change demonstrated greater pro-environmental engagement compared to those who lacked supportive coping resources (Ojala & Bengtsson, 2019). Similarly, Stevenson et al. (2019) found that adolescents who discussed climate concerns with friends and family reported stronger climate-related beliefs and greater intention to engage in PEB. These findings suggest that coping mechanisms, particularly those reinforced through supportive social contexts, play a critical role in determining whether CCA functions as a catalyst for action or a barrier to sustained engagement.

Climate Change Related Emotions and Pro-environmental Behavior

Translating concern about climate change into behavioral engagement remains a key question in understanding responses to environmental issues. Research suggests that individuals may be more likely to report engagement in PEBs, including both private and public actions, when they perceive climate change as personally and collectively relevant (Valdez et al., 2018).. When climate change is experienced as immediate and personally meaningful, it may become more emotionally salient, which can be associated with greater reported engagement (Valdez et al., 2018). Similarly, Stevenson et al. (2019) found that adolescents who recognize the risks of climate change report stronger pro-environmental intentions, defined as self-reported willingness or plans to engage in environmentally relevant behaviors. Perceptions of collective vulnerability, such as concern for community or national impacts, are also associated with greater support for climate-related policies and practices aimed at reducing environmental harm (Leiserowitz et al., 2021). Together, these findings suggest that climate-related emotions may be associated with increased behavioral engagement. However, emotional concern alone does not guarantee action, as individuals may experience inaction due to feelings of powerlessness and uncertainty (Wang

et al., 2025). Perceived environmental efficacy, defined here as the belief that one's individual or collective actions can contribute to environmental outcomes, may help explain when concern is associated with engagement. When individuals perceive that their efforts, particularly in combination with others, can contribute to broader change, they may be more likely to report sustained engagement in PEBs.

Climate change optimism (CCO) represents another emotional pathway through which climate-related concern may be associated with pro-environmental engagement. In the present study, CCO is conceptualized as a daily, self-reported belief that climate-related problems can be mitigated and that individual or collective actions can contribute to positive outcomes.

Individuals who report higher levels of climate-related optimism may be more likely to report engagement in PEBs and support for broader climate-related actions (Ojala & Bengtsson, 2019).

Exposure to optimistic perspectives in conversations may foster a sense of hope and encourage problem-solving, potentially counteracting disengagement associated with distress (Ojala & Bengtsson, 2019; Ojala & Lakew, 2017). However, the relationship between optimism and behavior is not necessarily linear, and high levels of optimism do not always translate into increased engagement (Ojala & Bengtsson, 2019). This highlights the importance of examining how optimism interacts with other emotional responses, such as anxiety, in shaping behavioral outcomes.

In contrast, climate change-related distress may be associated with reduced behavioral engagement. While moderate concern can increase attention to climate-related issues, higher levels of anxiety have been linked to avoidance, withdrawal from discussions, and lower reported engagement in PEBs (Wang et al., 2025). Feelings of hopelessness and emotional exhaustion may overwhelm individuals' sense of agency, reducing their likelihood of sustained

involvement in climate-related behaviors. Meaning, climate-related emotions may be associated with either increased or decreased engagement depending on their intensity and individuals' perceived capacity to respond.

The emotional responses elicited by climate change are complex, with stress and anxiety capable of producing both adaptive and maladaptive behavioral outcomes. Moderate levels of stress may act as a catalyst for action by heightening perceived urgency and activating problem-focused coping strategies that emphasize agency. In contrast, excessive anxiety may overwhelm an individual's coping resources, leading to emotional exhaustion, avoidance, or disengagement. Thus, the intensity and regulation of climate-related emotions play a critical role in determining whether individuals sustain or withdraw from pro-environmental engagement. Importantly, these emotional responses do not occur in isolation but are often shaped through social interactions, including conversations with others about climate change.

Climate Change Conversations: Emotional and Behavioral Consequences

Emotional experiences are often shaped through social interaction. Emotion contagion theory suggests that individuals may “catch” or mirror the emotions expressed by others during interpersonal exchanges, leading emotional states to spread across social networks (Hatfield et al., 1993). In the context of climate change discussions, exposure to another person's anxiety or optimism may therefore influence how individuals interpret and emotionally respond to climate-related threats. At the same time, social support has long been recognized as a critical protective factor in the face of stress. According to the Stress-Buffering Theory of Social Support, supportive social interactions can mitigate the negative psychological effects of stress by providing emotional validation, cognitive reframing, and coping resources (Cohen & Wills,

1985). In the context of CCA, conversations with others may therefore shape whether distress becomes overwhelming or translates into adaptive engagement.

Social support plays a vital role in psychological well-being by offering emotional validation, buffering stress, and fostering resilience (Pearce et al., 2020). Conversations that validate emotions while also emphasizing problem-solving may support adaptive coping and a sense of collective efficacy (Pearce et al., 2020). For example, Ojala and Bengtsson (2019) found that young people who discussed climate concerns with supportive peers reported lower levels of distress and greater emotional resilience. Emotional validation from peers may help reduce feelings of helplessness and support more constructive emotional responses. At the same time, the benefits of these interactions may depend on the nature of the discussion. While agreement and shared perspectives can provide emotional comfort, exposure to differing viewpoints may also encourage critical thinking and deeper engagement with the issue. Similarly, Latkin et al. (2024) found that individuals who engaged in climate change conversations reported lower feelings of isolation and higher levels of engagement, although conversations perceived as dismissive or invalidating were associated with frustration and withdrawal. Together, these findings suggest that peer conversations may either support or challenge individuals' emotional responses, depending on how those interactions are experienced. Both emotional validation and constructive dialogue may play important roles in shaping how individuals process and respond to climate-related concerns.

Beyond emotional outcomes, climate conversations may also be associated with behavioral engagement. Social influence research demonstrates that individuals' attitudes and behaviors are shaped through communication within their social networks (Stevenson et al., 2019). Family and peer discussions that reinforce concern and collective responsibility are

associated with higher levels of PEB, including both private actions and public actions (Dayton et al., 2023; Tian et al., 2023). A recent systematic review further found that interpersonal environmental communication is associated with a range of pro-environmental outcomes across age groups and contexts, including self-reported behaviors, behavioral intentions, and support for environmental policies (Tian et al., 2025). These findings suggest that conversations may not only be associated with emotional responses to climate change but also with broader patterns of behavioral engagement.

However, much of the existing literature treats communication as uniformly beneficial and often focuses on structured interventions rather than naturalistic peer exchanges. Less is known about how the emotional tone of climate conversations, such as expressions of anxiety, optimism, or disengagement, shapes daily fluctuations in CCA and PEB. Thus, while prior research establishes that peer conversations matter for both emotional well-being and environmental action, important questions remain regarding how the emotional outlook of conversation partners influences whether CCA is buffered or amplified over time.

The Present Study

This study examines how peer discussions influence climate change anxiety (CCA), climate change optimism (CCO), and PEBs at the daily level using a daily diary design. By tracking participants' emotional responses to climate-related conversations over time, this research aims to clarify how both individual emotional states and the perceived emotional tone of conversation partners shape climate-related experiences and behavioral engagement.

Specifically, this study investigates whether engaging in climate change conversations is associated with fluctuations in CCA, CCO, and PEBs, and whether the perceived emotional outlook of conversation partners, particularly their expressions of anxiety and optimism about

climate change, is associated with individuals' own emotional responses and behaviors. Drawing on emotion contagion theory and stress-buffering frameworks, this study examines whether peer interactions function as a source of emotional amplification or emotional support in the context of climate change.

It is first hypothesized that individuals' own emotional responses to climate change will be associated with PEB. Specifically, higher levels of CCA are expected to be positively associated with greater engagement in PEB (H1), reflecting the idea that concern about climate change may motivate action. Similarly, higher levels of CCO are expected to be positively associated with PEB (H2), as beliefs in the possibility of positive change and the effectiveness of individual or collective action may promote engagement. Although prior research suggests that very high levels of optimism may be associated with reduced urgency or complacency, the present study focuses on the general association between optimism and engagement.

In addition to these main effects, it is hypothesized that CCO will moderate the relationship between CCA and PEB (H3). Specifically, the positive association between anxiety and PEB is expected to be stronger at higher levels of optimism, suggesting that optimism may help channel climate-related concern into constructive action rather than disengagement.

This study also examined the role of peer emotional influence. Consistent with emotion contagion theory, it is expected that perceived peer emotional expressions may be associated with individuals' own emotional responses and behaviors. Specifically, perceived peer CCA is examined in relation to individuals' own CCA (H4), and perceived peer climate optimism is examined in relation to individuals' own optimism and PEB (H5a-H5b). The relationship between perceived peer optimism and individuals' own climate anxiety is explored without a directional prediction, given mixed findings in prior research (h5c)/

Together, these hypotheses aim to clarify how both individual emotional processes and interpersonal dynamics shape CCA and PEB in everyday life.

Method

Participants

Participants were undergraduate students at CUNY Brooklyn College recruited through the psychology department research participant pool and compensated with research participation credit applicable to psychology course requirements or extra credit.

Procedure

Participants voluntarily signed up for the study through the psychology department's research participation system. Participants attended an initial lab session, where they received an overview of the study and provided informed consent. Upon enrollment, participants were asked to complete a series of surveys over eight days to capture a range of psychological and social experiences during their transition to college life.

All data were collected online through Qualtrics, a secure web-based survey platform. The study began with a baseline survey that gathered demographic information and initial measures of social functioning and emotional well-being. Participants then completed an 8-day daily diary, during which they reported on their day-to-day social interactions, emotional experiences, and cognitive appraisals. Finally, participants completed a concluding survey. For the present thesis, data from the daily diaries were used to assess CCA, CCO, the occurrence of climate change conversations, and PEB for each of the 8 days of the diary period.

Materials

Baseline Survey

At the beginning of the study, participants completed a baseline survey assessing climate-related beliefs, emotional responses, and PEBs, along with additional measures not central to the present analyses. Participants were informed that the survey would focus on their views and behaviors related to climate change and were instructed to respond honestly. All responses were confidential. The baseline survey included the following measures:

Climate Distress. Climate distress was assessed using a seven-item scale adapted from Reser et al. (2012). Items measured emotional reactions to the threat of climate change, including worry, distress, guilt, and perceived personal responsibility (e.g., “I feel some sense of loss because of climate change impacts that are becoming apparent in my local area”). Responses were recorded on a 6-point scale ranging from 1 (Strongly disagree) to 6 (Strongly agree). The measure demonstrated good reliability ($\alpha = .914$).

Pro-Environmental Behaviors. Participants reported their engagement in private and public PEBs. Participants indicated how frequently each item applied to them on a 5-point scale ranging from 1 (Never) to 5 (Very often). Private PEB items included: waiting until having a full load before using the washing machine or dishwasher; preferring to travel by car even when public transportation is available (reverse-scored); turning off electronic devices and appliances when not in use; reducing heating or air conditioning use; limiting time spent in the shower; recycling; reusing plastic bags; turning off lights when leaving a room; purchasing locally produced food; buying sustainably packaged products; composting food waste; avoiding companies with poor environmental records; and choosing products that are less damaging to the environment when given the option. Reverse-worded items were reverse-scored prior to analysis, and items were averaged such that higher scores indicated greater engagement in private PEBs. The scale demonstrated acceptable internal consistency ($\alpha = .763$)

Public PEBs were assessed with 12 items reflecting outward-facing or socially engaged environmental behaviors. Participants responded using the same 5-point frequency scale. Items included: contributing time or resource to environmental organizations; educating oneself about environmental issues; contacting politicians or government officials; attending protests or marches; participating in conversation efforts; making voting decisions based on environmental policies; allocating time to focus on environmental issues; organizing environmental events; discussing environmental issues with others in one's social network; contributing to community sustainability efforts; using online platforms to raise awareness; and paying close attention to environmental issues in the media. Items were averaged such that higher scores indicated greater engagement in public PEBs. This scale demonstrated excellent internal consistency ($\alpha = .909$).

Climate Change Beliefs. Beliefs about climate change were measured using items adapted from the Yale Program of Climate Change Communication (YCOM). These four items assessed belief in the reality of climate change, perceived human causation, perceived scientific consensus, and beliefs about climate change impacts on weather patterns in the United States.

Six Americas Short Survey (SASSY). Participants completed the SASSY, a four-item audience segmentation instrument that classifies individuals based on their climate change views in terms of perceived importance, worry, expected personal harm, and expected harm to future generations. The four items cover: (1) how important climate change is personally; (2) how worried the respondent is about climate change; (3) how much harm they expect climate change to cause them personally; and (4) how much harm they expect climate change to cause future generations. Respondents choose from multiple-choice categories for each item (e.g., importance from *Not at all important* to *Extremely important*). Using established scoring algorithms, these four item responses can be entered into a group scoring tool to assign each respondent to one of

the Sic Americas climate opinion segments (Alarmed, Concerned, Cautious, Disengaged, Doubtful, Dismissive), allowing comparison between the study sample and broader public opinion patterns.

Additional Climate Change Questions. Exploratory items assessed perceived personal impact of climate change, self-rated climate knowledge, typical emotional reactions to climate change (e.g., anxiety, hope, anger), frequency of climate-related discussions with close and non-close others, perceived agreement with close others' beliefs, and encouragement of PEBs.

Daily Diary Survey

Participants completed a daily diary survey each evening for eight consecutive days. Daily surveys assessed participants' emotional responses to climate change, PEBs, and social interactions related to climate change. Participants were reminded to respond honestly, and confidentiality was emphasized. The daily diary included the following measures:

Daily Climate Change Anxiety (CCA). Daily CCA was assessed with one item: *"I felt anxious, distressed, or worried about climate change today."* Responses ranged from 1 (Not at all) to 7 (Extremely).

Daily Climate Change Optimism (CCO). Daily optimism was assessed with one item: *"I felt optimistic, hopeful, or confident about our ability to reduce harms of climate change today."* Responses ranged from 1 (Not at all) to 7 (Extremely).

Daily Pro-Environmental Behavior (PEB). Daily PEB was measured using two items assessing private and public actions: (1) limiting personal consumption or energy use, and (2) engaging in a public or advocacy-related action (e.g., activism, raising awareness). Items were rated on a 1 (Not at all) to 7 (Extremely) scale. A composite daily PEB score was created by averaging the two items, with higher scores indicating greater engagement in PEB.

Climate Change Conversations. Participants reported whether they had engaged in any climate-related conversations in the past 24 hours (Yes/No). If yes, participants indicated whom they spoke with (e.g., family, friends, romantic partners, coworkers, online contacts), briefly described the conversation content, and selected descriptors characterizing the tone of the conversation (e.g., inspiring, anxiety-provoking, motivating).

Perceived Peer Climate Emotions. For days when a conversation occurred, participants rated the perceived emotional tone of their conversation partner(s), including perceived anxiety (*“To what extent did the person you spoke with seem anxious, distressed, or worried about climate change?”*) and perceived optimism (*“To what extent did the person you spoke with seem optimistic, hopeful, or confident about our ability to reduce harms of climate change?”*). Responses ranged from 1 (Not at all) to 7 (Extremely).

Encouragement of Pro-Environmental Behavior. Participants reported whether they encouraged close others to engage in PEBs and whether close others encouraged them to do so. When applicable, participants rated the perceived success of these encouragements on a scale from 1 (Not at all) to 7 (Completely).

Data Analytics Strategy

Data Preparation and Cleaning

The daily diary dataset was originally provided as a .sav (SPSS) file and imported into R using the haven package. Initial inspection of the dataset included examining variable names, summary statistics, and the structure of the data to confirm the number of participants and observations.

The diary study was designed to include responses across eight days. Therefore, diary entries occurring outside the intended range were removed to ensure analyses reflected the

correct sampling window. Specifically, observations were filtered to include only diary days 1 through 8. After cleaning, the final dataset included 867 daily observations from 119 participants.

Conversation occurrence about climate change was originally coded with two response options indicating whether a conversation occurred. To facilitate analysis, this variable was recoded into a binary format (0 = no conversation, 1 = conversation). Missing values were retained and handled using the default missing data procedures within multilevel modeling, which allows all available observations to contribute to model estimation.

Descriptive analyses were conducted to examine the distribution of key variables, including CCA, CCO, and PEBs. As shown in Table C1, variable means, standard deviations, and frequencies fell within expected ranges across diary days.

Table C1

Descriptive Statistics for Key Study Variables

Variable	M	SD	Min	Max	N
Climate Change Anxiety	1.85	1.34	1.00	7.00	867
Climate Change Optimism	2.09	1.46	1.00	7.00	866
Private Pro-Environmental Behavior	2.37	1.57	1.00	7.00	867
Public Pro-Environmental Behavior	1.51	1.06	1.00	7.00	867

Daily Diary Compliance

A total of 119 participants contributed diary responses. Of 952 possible diaries, 867 observations (91.1%) were included in the final analyses after removing entries outside the intended diary window and cases with missing values on variables relevant to specific models.

The number of completed diary entries was relatively consistent across the eight days of the study (Day 1: 109; Day 2: 112; Day 3: 108; Day 4: 112; Day 5: 107; Day 6: 113; Day 7: 106; Day 8: 107). Participants reported engaging in conversations about climate change on 8.9% ($n = 77$) of diary days.

Analytic Plan

Daily diary observations (Level 1) were nested within participants (Level 2). Therefore, multilevel models were conducted in RStudio using the lme4 and lmerTest packages. Multilevel modeling was used to account for the nonindependence of repeated observations within individuals and to estimate both within-person and between-person variability in outcomes across diary days.

All models specified random intercepts for participants to account for variation in baseline levels of climate-related emotions and behaviors across individuals. In addition, random slopes for conversation occurrence were included when possible to allow the effect of climate conversations to vary across participants. Models were estimated using restricted maximum likelihood estimation.

To examine whether engaging in climate change conversations was associated with daily climate-related feelings and behaviors, a series of multilevel models were conducted in which conversation occurrence (0 = no conversation, 1 = conversation) was entered as a Level 1 predictor of the relevant outcome. Separate models were estimated for each outcome variable to evaluate the association between climate conversations and (a) daily CCA, (b) daily CCO, (c) private PEB, and (d) public PEB.

In each model, the outcome variable was entered as a continuous dependent variable measured at the daily level. Conversation occurrence served as the primary predictor in the first

set of multilevel models. These models allowed us to test whether participants reported different levels of climate-related emotions and behaviors on days when they engaged in climate conversations compared to days when they did not.

To test whether participants' own climate-related emotions predicted their own PEBs, additional multilevel models were conducted with daily CCA and daily CCO entered as Level 1 predictors of daily PEB. Separate models were estimated for private and public PEB outcomes. To test whether CCO amplified association between CCA and PEB, an interaction term between daily CCA and daily CCO was also included in each model. These models allowed us to examine whether anxiety and optimism were uniquely associated with PEB, and whether the relationship between anxiety and behavior differed depending on participants' level of optimism.

Finally, a mediation analysis was conducted to examine whether daily CCA helped explain the association between engaging in climate conversations and private PEB. This analysis tested whether climate conversations were associated with greater private PEB indirectly through increases in daily climate anxiety.

A final set of multilevel models examined whether the perceived emotional state of conversation partners was associated with participants' own daily emotions and behaviors. Specifically, perceived partner climate anxiety and perceived partner climate optimism were entered as predictors of participants' daily climate anxiety, climate optimism, private PEB, and public PEB. These analyses were conducted using only diary days on which participants reported having a climate change conversation and rated their conversation partners' emotional state.

Additional descriptive analyses were conducted to summarize the distribution of key variables and to estimate the proportion of diary days on which climate conversations occurred.

Means and standard deviations were calculated for daily climate anxiety, climate optimism, and PEB variables to provide descriptive context for the multilevel analyses.

Results

Effects of Conversation Engagement

To test whether engaging in a climate change conversation affected climate-related feelings and behaviors, we ran a series of multilevel models in which engaging in a climate change conversation was entered as a predictor of each outcome variable. Results revealed that CCA scores for people who engaged in climate change-related conversations ($M = 2.68$) was significantly higher than for those who did not engage in climate-related conversations ($M = 1.76$), $b = .83$, $SE = .16$, $t(48.51) = 5.02$, $p < .001$. However, participants' climate optimism did not differ on days when they engaged in climate conversations ($M = 2.38$) compared to days when they did not ($M = 2.07$), $b = 0.33$, $SE = 0.18$, $t(45.33) = 1.86$, $p = .069$.

A similar model examined whether engaging in climate conversations predicted public and private PEB. Results indicated that participants reported significantly greater private PEB on days when they engaged in climate conversations ($M = 2.94$) compared to days when they did not ($M = 2.32$), $b = 0.41$, $SE = 0.17$, $t(45.92)$, $p = .021$. Similarly, participants reported greater public PEB on days when climate conversations occurred ($M = 1.86$) than on days when they did not ($M = 1.47$). However, this effect did not reach conventional levels of statistical significance, $b = 0.26$, $SE = 0.15$, $t(49.58) = 1.80$, $p = .078$. See Table C2.

Table C2

Mediation of the Association Between Climate Conversations and Private Pro-Environmental Behavior Through Climate Change Anxiety

Effect	Estimate	t	p	CI
Conversations → Climate Change Anxiety	0.83	5.02	< .001	
Climate Change Anxiety → Private PEB	0.35	9.74	< .001	
Indirect effect (Conversations → CCA → Private PEB)	0.43		< .001	[0.25, 0.63]
Direct effect (Conversations → Private PEB)	0.18		.349	[-0.19, 0.56]
Total effect	0.61		.004	[0.22, 1.01]

Effect of Own Emotions on PEBs

To test whether one's own emotions predicted one's own PEBs (H1 & H2), and whether CCO amplified the effect of CCA on PEB (H3), two multilevel models were estimated, one for private PEB and one for public PEB. In each model, CCA and CCO were entered as predictors, along with their interaction term. Results indicated that both CCA and CCO were significant unique predictors of private and public PEB. The interaction between CCA and CCO was not statistically significant for private PEB, but was significant for public PEB, indicating that the association between anxiety and PEB was stronger at higher levels of optimism. See Table C3 for full results.

Table C3

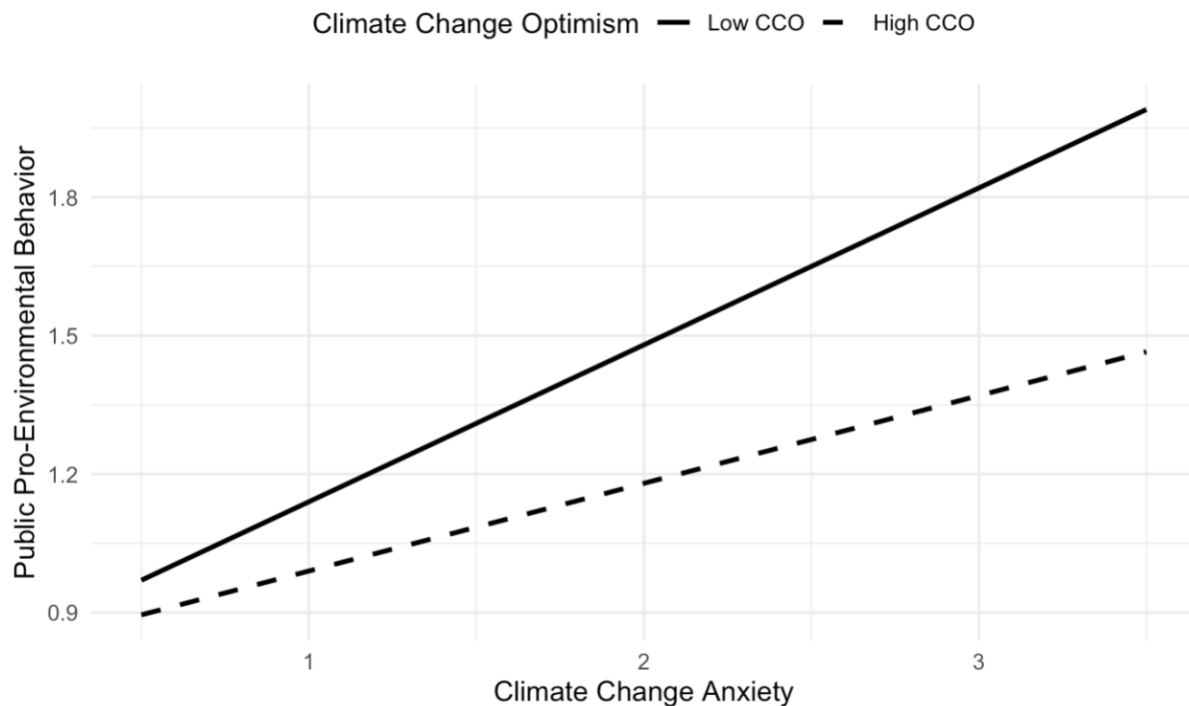
Multilevel Models Predicting Private and Public Pro-Environmental Behavior from Climate Change Anxiety, Climate Change Optimism, and Their Interaction

Outcome	Predictor	b	SE	df	t	p
Private PEB	Intercept	1.34	0.16	460.28	8.51	< .001
Private PEB	Climate Change Anxiety	0.16	0.06	832.87	2.60	0.009
Private PEB	Climate Change Optimism	0.31	0.06	860.76	5.37	< .001
Private PEB	CCA × CCO	0.02	0.02	838.02	1.19	0.233
Public PEB	Intercept	0.74	0.10	496.09	7.75	< .001
Public PEB	Climate Change Anxiety	0.15	0.04	855.07	3.73	< .001
Public PEB	Climate Change Optimism	0.11	0.04	819.74	2.86	0.004
Public PEB	CCA × CCO	0.05	0.01	858.67	4.20	< .001

The interaction between CCA and CCO in the prediction of public PEBs revealed that the positive effect of CCA on public PEBs was stronger at higher levels of CCO ($b = 0.34$, $se = 0.03$, $t=10.96$, $p<.001$) than at lower levels of CCO ($b = 0.19$, $se = 0.04$, $t=4.77$, $p<.001$), though the effect of CCA on public PEB was significant even at lower levels of CCO. See Figure C1.

Figure C1

Interaction between climate change anxiety and climate change optimism predicting public pro-environmental behavior



Test of Mediation

To examine whether CCA helped explain the association between climate conversations and private PEB, a mediation analysis was conducted (see Table 2). Conversation occurrence was entered as the predictor, daily CCA was entered as the mediator, and private PEB was entered as the outcome. This analysis tested whether engaging in climate conversations was indirectly associated with greater private PEB through increases in daily climate anxiety. The

indirect effect was estimated using a bootstrapped confidence interval approach, which is commonly used in mediation analyses because it does not assume a normal sampling distribution of the indirect effect. Results indicated a significant indirect effect of climate conversations on private PEB through CCA, $b = 0.43$, 95% $CI [0.25, 0.63]$, $p < .001$. The direct effect was not significant, $b = 0.18$, 95% $CI [-0.19, 0.56]$, $p = .349$, whereas the total effect was significant, $b = 0.61$, 95% $CI [0.22, 1.01]$, $p = .004$. This pattern suggests that the association between climate conversations and private PEB may be explained, in part, by increases in CCA. Although the direct effect was not statistically significant, the positive coefficient indicates that a direct association may still be present but was not detected in the current.

Effect of Partner's Emotional State

To examine whether the emotional tone of climate change conversations predicted participants' daily emotions and behaviors, a series of multilevel models was conducted in which perceived partner climate anxiety and perceived partner climate optimism were entered as predictors of each relevant outcome.

First, we examined whether perceived partner climate anxiety predicted participants' own climate-related emotions. Results indicated that perceived partner climate anxiety did not significantly predict participants' CCA, $b = 0.10$, $SE = 0.08$, $t(42.59) = 1.35$, $p = .183$. Similarly, perceived partner climate anxiety was not significantly associated with participants' CCO, $b = -0.05$, $SE = 0.06$, $t(36.87) = -0.73$, $p = .470$.

Next, we examined whether perceived partner climate anxiety predicted participants' PEBs. Results indicated that perceived partner climate anxiety did not significantly predict private PEB, $b = 0.07$, $SE = 0.11$, $t(73.94) = 0.65$, $p = .516$. Likewise, perceived partner climate

anxiety was not significantly associated with public PEB, $b = 0.02$, $SE = 0.08$, $t(62.38) = 0.31$, $p = .761$.

We next examined whether perceived partner climate optimism predicted participants' daily emotions. Results indicated that perceived partner climate optimism was not significantly associated with participants' CCA, $b = 0.10$, $SE = 0.09$, $t(49.33) = 1.06$, $p = .296$. Similarly, perceived partner climate optimism did not significantly predict participants' CCO, $b = 0.09$, $SE = 0.08$, $t(42.57) = 1.12$, $p = .268$.

Finally, we examined whether perceived partner climate optimism predicted PEBs. Results indicated that perceived partner climate optimism was positively associated with private PEB at a marginal level of statistical significance, $b = 0.24$, $SE = 0.13$, $t(72.01) = 1.92$, $p = .059$, suggesting a potential trend in which higher perceived partner optimism may be associated with greater engagement in private pro-environmental actions. However, perceived partner climate optimism was not significantly associated with public PEB, $b = -0.01$, $SE = 0.09$, $t(69.32) = -0.07$, $p = .944$.

Discussion

Overall, findings provided support for hypotheses related to individuals' own emotions and behavior, but limited support for hypotheses related to peer emotional influence. Using an eight-day daily diary design, participants reported their experiences and behaviors each day, allowing examination of within-person variation over time. Results from multilevel models indicated that engaging in conversations about climate change was associated with significantly higher levels of CCA and greater engagement in private PEBs. In contrast, conversations were not significantly associated with public PEB or climate optimism, although both outcomes were in the expected direction. Similarly, perceived partner anxiety and optimism were not

significantly associated with participants' daily emotions or public PEB, and perceived partner optimism was also not significantly associated with private PEB.

Climate Conversations and Climate Anxiety

Consistent with expectations, participants reported significantly higher CCA on days when they discussed climate change than on days when they did not. This finding suggests that conversations about climate change may heighten awareness of environmental threats or bring climate-related concerns to the forefront of individuals' attention. However, because the data are correlational, it is also possible that individuals who are already feeling more anxious about climate change are more likely to engage in conversations about the topic on those days. Prior research suggests that CCA is increasingly common, particularly among adolescents, and may influence emotional well-being and behavioral engagement with environmental issues (Clayton & Karazsia, 2020; Hickman et al., 2021). Conversations may involve discussing the potential impacts of climate change, sharing concerning information, or expressing worry about environmental outcomes, all of which may contribute to heightened anxiety or distress.

At the same time, this increase in climate anxiety may not necessarily represent a maladaptive response. Prior research suggests that moderate levels of climate-related concern may motivate individuals to think more seriously about environmental issues and potentially take action to mitigate environmental harm. Moderate levels of concern about climate change have been linked to greater motivation to engage in PEB and environmental advocacy (Clayton, 2020). Thus, experiencing heightened anxiety following climate conversations may reflect increased engagement with the issue rather than negative emotional outcomes. Consistent with this possibility, the present findings provide some evidence that climate-related anxiety may be linked to greater engagement in PEB.

Climate Conversations and Climate Optimism

In addition to climate anxiety, climate conversations were also associated with slightly higher levels of climate optimism, although this effect was not statistically significant. This pattern may reflect the mixed nature of climate-related discussions. Some conversations may emphasize solutions, collective action, or progress in addressing climate change, fostering optimism. Others may focus on the severity or urgency of climate threats, potentially increasing anxiety instead. As a result, the emotional impact of climate conversations may depend on the tone and content of the discussion.

Future research could explore the characteristics of climate conversations in more detail, including whether conversations are perceived as motivating, depressing, inspiring, or argumentative. Examining these qualitative aspects may help clarify when and how climate conversations influence emotional responses and behavior.

Climate Conversations and Pro-Environmental Behavior

In addition to emotional responses, climate conversations were associated with greater engagement in private PEBs. Participants reported more behaviors, such as limiting energy use or personal consumption, on days when they discussed climate change. This finding aligns with theories of social influence and social norms, which suggest that discussing an issue with others can reinforce its perceived importance and motivate individuals to act in ways that are consistent with shared values or concerns.

Interestingly, the relationship between climate conversations and public PEB did not reach statistical significance, although the trend was in a positive direction. Interpersonal discussions can shape environmental attitudes and behaviors by reinforcing social norms and increasing the perceived importance of environmental issues (Goldberg et al., 2020). One

possible explanation is that private PEBs are generally easier and require fewer resources or social risks than public actions such as activism or advocacy. While conversations may increase awareness and motivation, translating this motivation into public action may require additional structural or social support.

An important consideration when interpreting these findings is that the CCO measure may reflect not only optimism, but also perceived efficacy. The item assessed how optimistic, hopeful, or confident participants felt about the ability to reduce the harms of climate change. Because confidence in reducing harm overlaps conceptually with efficacy, this measure may capture beliefs about whether individual or collective action can make a difference. This distinction may help explain why optimism was associated with greater PEB and why it strengthened the relationship between anxiety and public PEB. In this context, it may not be optimism alone that motivates action, but rather a sense that action is meaningful and effective.

Strengths and Contributions

The present study offers several important contributions. First, the use of a daily diary design enabled examination of within-person fluctuations in climate conversations, emotions, and behaviors over time, providing a more ecologically valid understanding of how these processes unfold in everyday life. This approach extends prior research that has largely relied on cross-sectional methods by capturing day-to-day variability in experiences and responses. Second, the study assessed naturally occurring climate conversations by allowing participants to report on interactions with a range of conversation partners, rather than restricting discussions to specific relationship types. This design increases the generalizability of the findings and better reflects the diverse social contexts in which climate-related discussions occur. Finally, by examining both emotional responses and behavioral outcomes, the study provides a more

comprehensive perspective on how interpersonal communication may shape engagement with climate change.

Partner Emotional Tone of Conversations

The present study also examined whether the perceived emotional state of conversation partners was associated with participants' own emotions and behaviors. Overall, perceived partner anxiety and optimism were not significantly associated with participants' daily climate anxiety, climate optimism, or public PEB. However, perceived partner climate optimism was not significantly associated with private PEB, although the effect was in the expected direction. Although this effect did not reach conventional levels of statistical significance, it suggests that conversations in which others express hope or confidence about addressing climate change may encourage individuals to engage in small, everyday environmental actions. Although not statistically significant, this pattern is consistent with theories of emotional contagion, which suggest that individuals may adopt or mirror the emotional tone expressed by others during social interactions (Hatfield et al., 1993). Optimistic conversations may therefore help reinforce perceptions that individual actions can make a difference, potentially motivating engagement in private environmental behaviors.

Limitations

Several limitations should be considered when interpreting the findings of this study. First, the study relied on self-reported measures of climate-related emotions, behaviors, and conversation characteristics. Although the daily diary design helps reduce recall bias by capturing experiences close in time to when they occur, self-report data may still be subject to social desirability effects or inaccuracies in reporting. Future research could incorporate more objective behavioral measures, such as tracking actual energy use, transportation choices, or

participation in environmental initiatives, to further validate these findings. Researchers could also assess climate conversations more objectively by, for example, recording conversations (with participants' consent) and coding their emotional tone, content, or level of conflict using trained independent observers. Additionally, ecological momentary assessment methods, such as prompting participants to report their emotions and behaviors multiple times throughout the day via mobile devices, could provide more real-time data and further reduce retrospective bias.

Second, the diary study spanned only eight days, which may limit the ability to capture longer-term patterns in climate conversations and behavioral responses. Climate-related attitudes and behaviors may fluctuate over longer time periods or in response to major environmental events or media coverage. For example, events such as extreme weather (e.g., wildfires, hurricanes, or heatwaves) or increased media attention to climate change may temporarily heighten concern, increase climate-related conversations, and motivate greater engagement in PEBs. Conversely, periods with less media attention or fewer salient environmental events may be associated with lower levels of concern and engagement. In addition, climate change conversations occurred on a relatively small proportion of diary days in the present study, which may have reduced statistical power and made it more difficult to detect smaller or more nuanced effects, particularly in analyses restricted to conversation days. Extending the diary period or conducting longer longitudinal studies would allow researchers to better capture how these broader contextual factors shape fluctuations in climate-related emotions and behaviors over time while also increasing the likelihood of observing a greater number of climate-related conversations.

Third, another important consideration is the potential for demand characteristics in the daily diary design. Because participants were prompted to reflect on CCA, conversations, and

behaviors each day, it is possible that the study itself increased the salience of climate change. This repeated prompting may have influenced whether participants noticed, initiated, or reported climate-related conversations during the study period. As a result, the observed patterns may partially reflect responses to the study context rather than entirely naturalistic behavior. Future research could examine whether these patterns persist in less structured settings or assess changes across diary days to determine whether repeated reflection influences emotions and behavioral responses over time.

Fourth, while multilevel modeling allowed us to examine within-person associations across diary days, the correlational nature of the data prevents strong causal conclusions. It is possible that individuals who were already feeling more anxious about climate change were more likely to initiate or engage in climate conversations on those days. Experimental or intervention-based studies would help clarify the causal relationship between climate conversations, emotional responses, and behavioral outcomes. For example, researchers could experimentally assign participants to engage in a structured conversation about climate change that varies in emotional tone, such as one emphasizing threat and concern versus one emphasizing solutions, efficacy, and collective action, and then assess subsequent changes in climate anxiety, optimism, and PEB. Intervention studies could also examine whether communication-based programs that encourage constructive climate dialogue increase engagement while reducing the risk of overwhelming anxiety.

Fifth, the sample may not be fully representative of the broader population, which may limit the generalizability of the findings. Participants who are willing to take part in a study about climate-related experiences may already be more engaged with environmental issues, more attentive to climate-related emotions, or more likely to discuss climate change in their daily lives

than the general public. As a result, the observed associations between climate conversations, emotions, and PEB may be stronger in this sample than they would be among less engaged individuals. For people who are less concerned about climate change or less involved in environmental issues, climate conversations may occur less often, evoke weaker emotional responses, or be less likely to translate into behavioral change. At the same time, it is also possible that conversations could have a different kind of impact for less engaged individuals by increasing awareness of climate change in ways that make the issue feel more personally relevant. Future research should examine whether these patterns differ across people with varying levels of baseline environmental concern and involvement.

Sixth, it is important to note that the present study was designed to examine associations between climate-related emotions, social interactions, and self-reported behaviors, rather than to promote specific behavioral outcomes. As such, the findings should be interpreted as descriptive of psychological processes rather than prescriptive of particular forms of engagement. While the analyses focused on PEB, the study does not evaluate the broader effectiveness or desirability of these behaviors, nor does it assume that increased engagement is inherently beneficial across all contexts.

In addition, the study did not include measures of general anxiety or non-climate-related daily activity, which limits the ability to determine whether the observed associations are specific to climate-related processes. For example, it is possible that days characterized by increased activity or social interaction more broadly may be associated with both greater anxiety and higher reported engagement in various behaviors. Without comparison to non-climate-related experiences, it is difficult to fully isolate climate change as the primary driver of these patterns. Future research could address this limitation by including measures of general emotional states

and non-climate-related behaviors to better distinguish climate-specific effects from broader daily fluctuations.

Implications and Future Directions

Research suggests that interpersonal climate conversations play an important role in shaping public engagement with climate change and encouraging collective action (Goldberg et al., 2020). Given the increasing prevalence of CCA among young people, this study offers valuable insights into how interpersonal discussions can either serve as a protective mechanism that fosters resilience and engagement or as a risk factor that amplifies anxiety and discourages action. The present findings suggest that everyday conversations about climate change may play an important role in shaping both emotional responses and behavioral engagement.

By increasing awareness and reinforcing the importance of environmental issues, conversations may serve as a mechanism through which individuals become more motivated to engage in PEBs. At the same time, the association between conversations and heightened anxiety highlights the importance of how climate issues are discussed. Conversations that emphasize collective solutions, efficacy, and shared responsibility may help channel concern into constructive engagement rather than overwhelm.

Future research could build on these findings by examining climate conversations in greater detail, including how specific conversation partners, the intensity or valence of emotional tone, and the particular topics discussed shape emotional and behavioral outcomes. Although the present study assessed with whom participants talked and whether partners expressed climate anxiety or optimism, future work could investigate these dimensions more comprehensively by distinguishing among different relationship types, assessing whether conversations are perceived as supportive or conflictual, and identifying whether discussions focus on solutions, personal

impacts, politics, or environmental loss. Understanding these more nuanced features of climate conversations may help inform communication strategies to promote constructive engagement with climate change.

Overall, the findings highlight the potential importance of interpersonal communication in shaping climate-related attitudes and actions. Encouraging open and constructive dialogue about climate change may be a pathway to greater awareness and engagement with environmental issues.

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Appendix

Appendix A

Baseline Survey Measures

Climate Distress

The following instructions were provided to participants: *The next set of questions is about climate change. These questions address your climate-related beliefs, opinions, and behavior. Again, please answer these questions as honestly as you are able. Once again, all of your responses are confidential.*

Participants responded to the following items:

Some people may find the threat of climate change to be particularly distressing. This may not be the case for you, and it is important that you respond in the context of your own personal experience and feelings. Please indicate the extent to which each of the following statements reflects your own response to the threat of climate change.

Responses were made on a 6-point Likert-type scale ranging from 1 (Strongly disagree) to 6 (Strongly agree).

1. I experience some distress each time I see or read media coverage of the likely impacts and consequences of climate change.
2. At times, I find myself thinking about and worrying about what the world will really be like for future generations because of climate change.

3. I experience some guilt over the fact that my family and friends' lifestyles and consumption patterns are in part responsible for the unfolding impacts of climate change.
4. It upsets me that there seems to be so little that I can do to address environmental problems such as climate change.
5. At times, I feel some personal responsibility for the problems and unfolding impacts of climate change.
6. The threat of climate change is affecting my quality of life and my assessment of environmental quality more generally.
7. I feel some sense of loss because of climate change impacts that are becoming apparent in my local area.

Private Pro-Environmental Behaviors

Participants were instructed: *Consider the following statements about environmental behaviors and indicate how frequently each item applies to you.*

Responses were made on a 5-point Likert-type scale ranging from 1 (Never) to 5 (Very often).

1. I wait until I have a full load before using the washing machine or dishwasher.
2. Even when public transportation options are available, I prefer to travel by car.
3. I make sure to turn off electronic devices and appliances when they're not in use.
4. I cut down on heating or air conditioning to decrease my use of energy.
5. I limit time spent in the shower in order to conserve water.
6. I recycle everything that I can.
7. I save and reuse plastic bags.
8. I turn off lights when I leave the room.
9. I prefer to buy food that I know is produced close to home.

10. I make an effort to buy products that are packaged sustainably.
11. I compost my food waste.
12. I avoid supporting companies that have a poor environmental record.
13. When I have a choice between products, I purchase the one that is less damaging to the environment.

Public Pro-Environmental Behaviors

Participants were instructed: *Consider the following statements about environmental behaviors and indicate how frequently each item applies to you.*

Responses were made on a 5-point Likert-type scale ranging from 1 (Never) to 5 (Very often).

1. I contribute resources or time to environmental organizations.
2. I make an effort to educate myself about environmental issues (e.g., through media, television, internet, etc.)
3. I reach out to politicians or other government officials about their stance on environmental issues.
4. I attend protests or marches related to climate change or other environmental concerns.
5. I take part in efforts to conserve nature and/or threatened wildlife.
6. I make voting decisions based on a candidate's environmental policies.
7. I consciously make time in my schedule to focus on environmental issues.
8. I get involved in organizing protests, workshops, or other events that have an environmental focus.
9. I speak to people in my social network (e.g., friends, family, partners, coworkers) about climate change or other environmental issues.
10. I contribute to projects that work toward community-level sustainability.

11. I use online tools (e.g., social media, YouTube, discussion forums) to raise awareness about environmental issues.
12. I pay close attention when climate change and other environmental issues are discussed in the media.

Climate Change Beliefs (YCOM)

Participants responded to the following items:

1. On a scale from 1 to 7, how strongly do you believe that climate change is or is not happening?
 - 1 = I strongly believe that climate change is NOT happening
 - 4 = I am unsure whether or not climate change is happening
 - 7 = I strongly believe climate change IS happening
2. Assuming climate change *is* happening: How much of it do you believe is caused by human activities, natural changes in the environment, or some combination of both?
 - 1 = I believe climate change is caused entirely by natural changes in the environment
 - 4 = I believe climate change is caused equally by natural changes and human activities
 - 7 = I believe climate change is caused entirely by human activities
3. To the best of your knowledge, what percentage of climate scientists have concluded that human-caused climate change is happening?
 - Responses range: 0% to 100%
4. How strongly do you agree or disagree with this statement: *Climate change is affecting the weather in the United States.*

- 1 = Strongly disagree
- 2 = Somewhat disagree
- 3 = Somewhat agree
- 4 = Strongly agree

Six Americas Short Survey (SASSY)

Participants responded to the following items:

1. How important is the issue of climate change to you personally?

- 1 = Not at all important
- 2 = Not too important
- 3 = Somewhat important
- 4 = Very important
- 5 = Extremely important

2. How worried are you about climate change?

- 1 = Not at all worried
- 2 = Not very worried
- 3 = Somewhat worried
- 4 = Very worried

3. How much do you think climate change will harm you personally?

- 1 = Not at all
- 2 = Only a little
- 3 = A moderate amount
- 4 = A great deal
- 5 = Don't know

4. How much do you think climate change will harm future generations of people?

- ☐ 1 = Not at all
- ☐ 2 = Only a little
- ☐ 3 = A moderate amount
- ☐ 4 = A great deal
- ☐ 5 = Don't know

Additional Climate Change Questions

1. To what extent do you feel you have been directly impacted by the effects of climate change?

- ☐ 1 = Not at all
- ☐ 4 = A moderate amount
- ☐ 7 = Severe impact

2. How would you assess your own level of knowledge about climate change (e.g., causes, consequences, and projected scenarios)?

- ☐ 1 = No knowledge
- ☐ 4 = Moderate knowledge
- ☐ 7 = Highly knowledgeable

3. When you think about the issues of climate change, to what extent do you usually feel:
(Responses were made on a 7-point scale ranging from 1 (Not at all) to 7 (Extremely))

- ☐ Depressed
- ☐ Anxious
- ☐ Hopeful
- ☐ Angry

- Determined
 - Scared
4. How frequently do you typically discuss climate change with close others (e.g., family, close friends, romantic partners)?
- 1 = Never
 - 2 = Once a month or less
 - 3 = A few times a month
 - 4 = Once a week or more
 - 5 = Almost every day
5. How frequently do you typically discuss climate change with non-close others (e.g., acquaintances, coworkers, online contacts)?
- 1 = Never
 - 2 = Once a month or less
 - 3 = A few times a month
 - 4 = Once a week or more
 - 5 = Almost every day
6. To what extent do you feel your beliefs about climate change are shared by your close others (e.g., family, close friends, romantic partners)?
- 1 = Everybody disagrees with me
 - 2 = The majority disagrees with me
 - 3 = Half and half
 - 4 = The majority agrees with me
 - 5 = Everybody agrees with me

7. How frequently do you encourage your close others to engage in pro-environmental behaviors?

- 1 = Never
- 2 = Rarely
- 3 = Sometimes
- 4 = Often
- 5 = Very often

8. How frequently do your close others encourage you to engage in pro-environmental behaviors?

- 1 = Never
- 2 = Rarely
- 3 = Sometimes
- 4 = Often
- 5 = Very often

Appendix B

Daily Diary Survey Measures

Participants completed a daily diary survey assessing their climate-related emotions, behaviors, and conversations from that day.

Daily Climate-Related Emotions and Behaviors

Participants were instructed: *Please answer the following questions about your feelings and actions around climate change TODAY.*

Responses to the following items were made on a 7-point Likert-type scale ranging from 1 (Not at all) to 7 (Extremely).

1. I felt anxious, distressed, or worried about climate change.
2. I felt optimistic, hopeful, or confident about our ability to reduce harms of climate change.
3. I limited my personal consumption and/or energy use in order to lessen my impact on the environment.
4. I made an effort to fight against climate change publicly (e.g., through activism, education, or using online tools to raise awareness).

Daily Climate Conversation Occurrence

Participants indicated whether they had engaged in a climate-related conversation in the past 24 hours.

1. Did you have any conversations about climate change in the last 24 hours?
 - Yes
 - No

If participants responded *Yes*, they completed the following follow-up items.

Daily Climate Conversation Characteristics

1. Who did you speak to about climate change? (Select all that apply.)
 - Immediate family (e.g., parents, siblings)
 - Extended family (e.g., aunts, uncles, cousins, nieces, nephews)
 - Romantic partner
 - Close friends
 - Acquaintances

- Colleagues/coworkers
 - Classmates
 - Neighbors
 - Online contacts (e.g., social media)
 - Other (please specify)
- 2. In just a few words, what did you talk about in your conversation(s) about climate change?
- 3. How would you characterize the conversation(s) you had with others about climate change (Select all that apply.)
 - Successful
 - Productive
 - Conflictual
 - Argumentative
 - Inspiring
 - Depressing
 - Anxiety provoking
 - Angering
 - Interesting
 - Motivating
 - Other (please specify)
- 4. To what extent did the person (or people) you spoke with seem anxious, distressed, or worried about climate change?

- Responses were made on a 7-point scale ranging from 1 (Not at all) to 7 (Extremely).
- 5. To what extent did the person (or people) you spoke with seem optimistic, hopeful, or confident about our ability to reduce harms of climate change?
 - Responses were made on a 7-point scale ranging from 1 (Not at all) to 7 (Extremely).

Daily Social Influence Around Pro-Environmental Behavior

1. Did you encourage any close others (e.g., family, close friends, romantic partners) to engage in pro-environmental behavior today?
 - Yes
 - No

If *Yes*:

1. How successful were you?
 - 1 = Not at all
 - 2 = Slightly
 - 3 = Moderately
 - 4 = Very
 - 5 = Completely
2. Did any of your close others (e.g., family, close friends, romantic partners) encourage you to engage in pro-environmental behavior today?
 - Yes
 - No

If *Yes*:

1. How successful were they?

- 1 = Not at all
- 2 = Slightly
- 3 = Moderately
- 4 = Very
- 5 = Completely

Appendix C

Supplementary Tables and Figures

Table C1

Descriptive Statistics for Key Study Variables

Variable	M	SD	Min	Max	N
Climate Change Anxiety	1.85	1.34	1.00	7.00	867
Climate Change Optimism	2.09	1.46	1.00	7.00	866
Private Pro-Environmental Behavior	2.37	1.57	1.00	7.00	867
Public Pro-Environmental Behavior	1.51	1.06	1.00	7.00	867

Table C2

Mediation of the Association Between Climate Conversations and Private Pro-Environmental Behavior Through Climate Change Anxiety

Effect	Estimate	t	p	CI
Conversations → Climate Change Anxiety	0.83	5.02	< .001	
Climate Change Anxiety → Private PEB	0.35	9.74	< .001	
Indirect effect (Conversations → CCA → Private PEB)	0.43		< .001	[0.25, 0.63]
Direct effect (Conversations → Private PEB)	0.18		.349	[-0.19, 0.56]
Total effect	0.61		.004	[0.22, 1.01]

Table C3

Multilevel Models Predicting Private and Public Pro-Environmental Behavior from Climate Change Anxiety, Climate Change Optimism, and Their Interaction

Outcome	Predictor	b	SE	df	t	p
Private PEB	Intercept	1.34	0.16	460.28	8.51	< .001
Private PEB	Climate Change Anxiety	0.16	0.06	832.87	2.60	0.009
Private PEB	Climate Change Optimism	0.31	0.06	860.76	5.37	< .001
Private PEB	CCA × CCO	0.02	0.02	838.02	1.19	0.233
Public PEB	Intercept	0.74	0.10	496.09	7.75	< .001
Public PEB	Climate Change Anxiety	0.15	0.04	855.07	3.73	< .001
Public PEB	Climate Change Optimism	0.11	0.04	819.74	2.86	0.004
Public PEB	CCA × CCO	0.05	0.01	858.67	4.20	< .001

Figure C1

Interaction Between Climate Change Anxiety and Climate Change Optimism Predicting Public Pro-Environmental Behavior

